



第四节 变系数方程

考虑变系数椭圆方程：

$$\frac{\partial}{\partial x} \left(a(x, y) \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(a(x, y) \frac{\partial u}{\partial y} \right) - c(x, y)u = f(x, y)$$

其中 $a(x, y) > 0$, $b(x, y) > 0$, $c(x, y) > 0$,

1：直接差分方法

$$\frac{1}{h^2} \delta_x (a_{ij} \delta_x u_{ij}) + \frac{1}{h^2} \delta_y (a_{ij} \delta_y u_{ij}) - c_{ij} u_{ij} = f_{ij}$$



2：有限体积法（积分差分方法）

$$\iint_{D_{ij}} \frac{\partial}{\partial x} \left(a(x, y) \frac{\partial u}{\partial x} \right) dx dy$$

$$= \int_{y_{j-\frac{1}{2}}}^{y_{j+\frac{1}{2}}} \left[a(x_{i+\frac{1}{2}}, y) \frac{\partial u}{\partial x} (x_{i+\frac{1}{2}}, y) - a(x_{i-\frac{1}{2}}, y) \frac{\partial u}{\partial x} (x_{i-\frac{1}{2}}, y) \right] dy$$

对上面定积分利用中点矩形公式有

$$\begin{aligned} \iint_{D_{ij}} \frac{\partial}{\partial x} \left(a(x, y) \frac{\partial u}{\partial x} \right) dx dy &\approx \left[a(x_{i+\frac{1}{2}}, y_j) \frac{\partial u}{\partial x} (x_{i+\frac{1}{2}}, y_j) \right. \\ &\quad \left. - a(x_{i-\frac{1}{2}}, y_j) \frac{\partial u}{\partial x} (x_{i-\frac{1}{2}}, y_j) \right] k \end{aligned}$$

$$\iint_{D_{ij}} \frac{\partial}{\partial x} (b(x, y) \frac{\partial u}{\partial x}) dx dy \approx [b(x_i, y_{j+\frac{1}{2}}) \frac{\partial u}{\partial x}(x_i, y_{j+\frac{1}{2}}) - b(x_i, y_{j-\frac{1}{2}}) \frac{\partial u}{\partial x}(x_i, y_{j-\frac{1}{2}})] h$$

$$\int_{D_{ij}} c(x, y) u dx dy \approx c_{ij} u_{ij} h k, \quad \int_{D_{ij}} f(x, y) dx dy \approx f_{ij} h k$$

于是有

$$\begin{aligned} & \frac{1}{h^2} [a_{i+\frac{1}{2}, j} (u_{i+1, j} - u_{i+1, j}) - a_{i-\frac{1}{2}, j} (u_{i, j} - u_{i-1, j})] \\ & + \frac{1}{h^2} [a_{i, j+\frac{1}{2}} (u_{i, j+1} - u_{i, j}) - a_{i, j-\frac{1}{2}} (u_{i, j} - u_{i, j-1})] - c_{ij} u_{ij} = f_{ij} \end{aligned}$$



第六节 特征值问题

The Eigenvalue Problem

Model Problem

Statement

Find nontrivial (u, λ) such that

$$-u_{xx} = \lambda u, \quad x \in (0, 1)$$

$$u(0) = u(1) = 0;$$

denote solutions (u^k, λ^k) , $k = 1, 2, \dots$, with

$$0 \leq \lambda^1 \leq \lambda^2 \leq \dots$$

N15

The Eigenvalue Problem

Exact Solution

$$-u_{xx} - \lambda u = 0$$

$\Downarrow$

$$u = A \sin \sqrt{\lambda} x + B \cos \sqrt{\lambda} x$$

$$u(0) = 0 \Rightarrow B = 0$$

$$u(1) = 0 \Rightarrow A = 0 \text{ or } \lambda = k^2 \pi^2, k = 1, 2, \dots$$

The Eigenvalue Problem

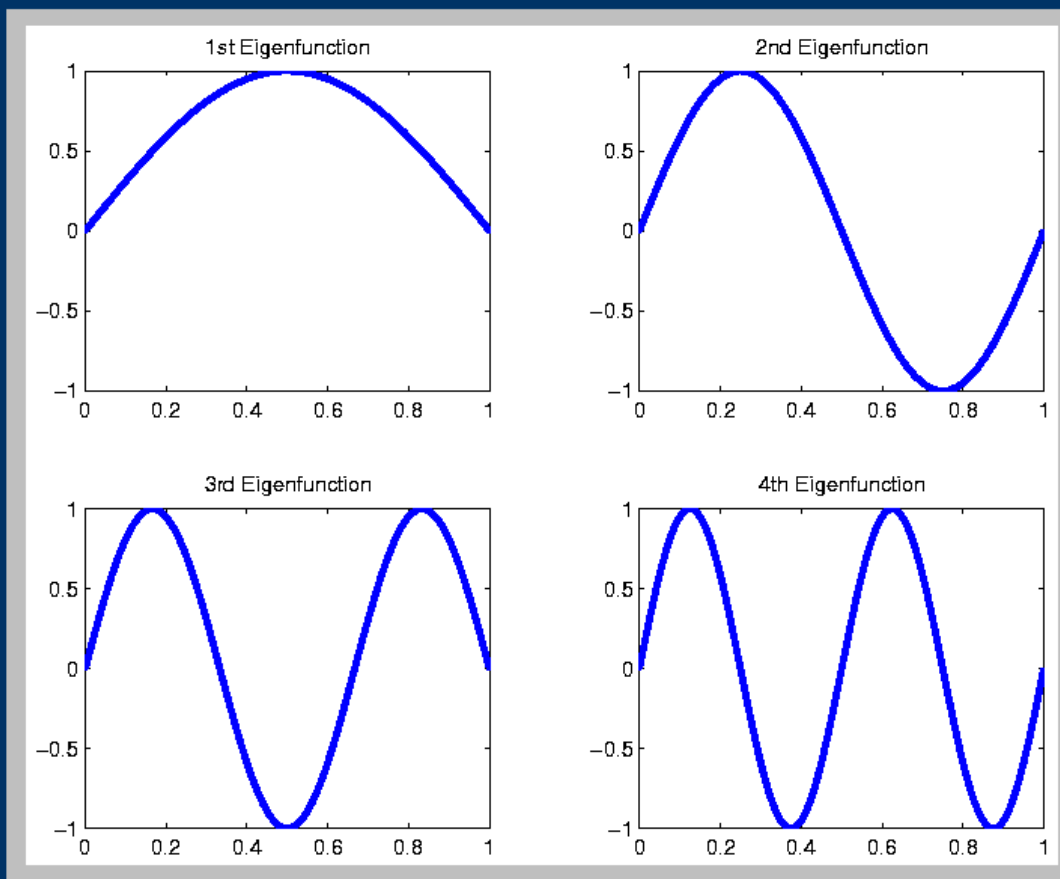
Thus (choose $A = 1$)

$$\left. \begin{array}{l} u^k = \sin k\pi x \\ \lambda^k = k^2\pi^2 \end{array} \right\} k = 1, 2, \dots$$

Larger $k \Rightarrow$ more oscillatory $u^k \Rightarrow$ larger λ .

The Eigenvalue Problem

Exact Solution

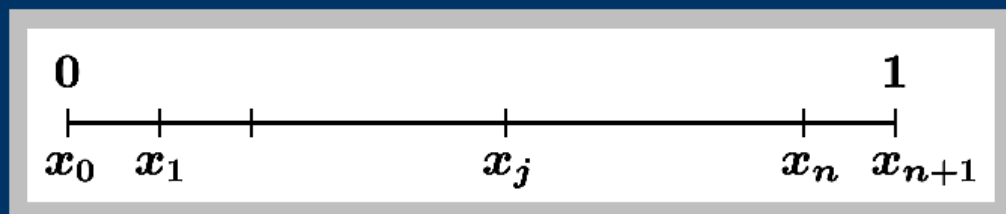


The Eigenvalue Problem

Discrete Equations

Difference Formulas

$$-u_{xx} = \lambda u, \quad u(0) = u(1) = 0$$



$$\Delta x = \frac{1}{n+1}$$

$$\frac{-1}{\Delta x^2}(\hat{u}_{j-1} - 2\hat{u}_j + \hat{u}_{j+1}) = \hat{\lambda}\hat{u}_j, \quad j = 1, \dots, n$$

$$\hat{u}_0 = \hat{u}_{n+1} = 0$$

The Eigenvalue Problem

Error Analysis

Analytical Solution: $\hat{\underline{u}}^k, \hat{\lambda}^k \dots$

Claim that

$$\hat{\underline{u}}^k \equiv \underline{u}^k$$

$$\begin{aligned}\hat{u}_j^k &= u^k(x_j) = \sin(k\pi x_j) \\ &= \sin(k\pi j \Delta x) = \sin\left(\frac{k\pi j}{n+1}\right), \quad j = 1, \dots, n\end{aligned}$$

Note $\hat{u}_0^k = \hat{u}_{n+1}^k = 0$ since $\sin(0) = \sin(k\pi) = 0$.

The Eigenvalue Problem

Error Analysis

...Analytical Solution: $\hat{u}^k, \hat{\lambda}^k$...

What are $\hat{\lambda}^k$?

$$-\frac{1}{\Delta x^2} \{ \hat{u}_{j-1}^k - 2\hat{u}_j^k + \hat{u}_{j+1}^k \}$$

$$= -\frac{1}{\Delta x^2} \{ \sin(k\pi(x_j - \Delta x)) - 2\sin(k\pi x_j) + \sin(k\pi(x_j + \Delta x)) \}$$

$$= -\frac{1}{\Delta x^2} \{ \underbrace{\sin(k\pi x_j - k\pi \Delta x) + \sin(k\pi x_j + k\pi \Delta x)}_{2 \cos(k\pi \Delta x) \sin(k\pi x_j)} - 2\sin(k\pi x_j) \}$$

The Eigenvalue Problem

Error Analysis

...Analytical Solution: $\underline{\hat{u}}^k, \hat{\lambda}^k$

Thus:

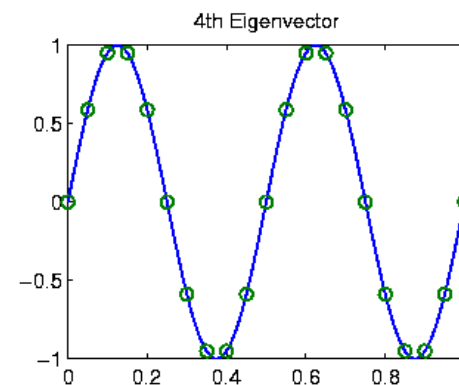
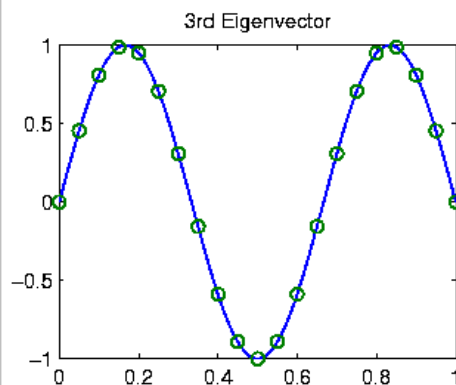
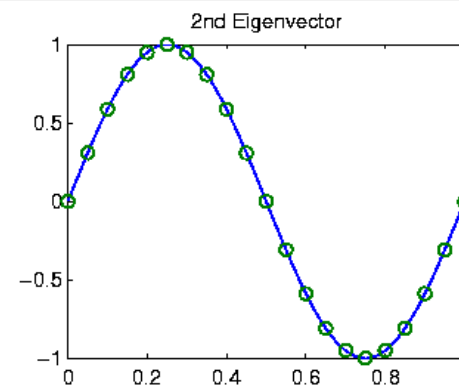
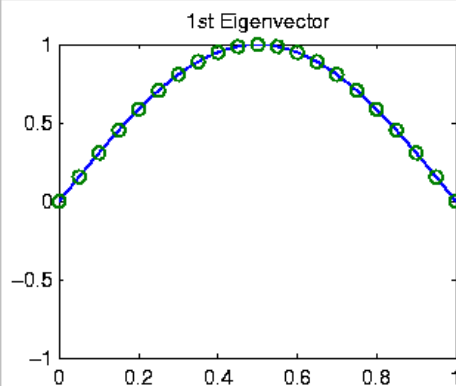
$$\begin{aligned} & -\frac{1}{\Delta x^2} \{ \hat{u}_{j-1}^k - 2\hat{u}_j^k + \hat{u}_{j+1}^k \} \\ &= -\frac{1}{\Delta x^2} \{ 2 \cos(k\pi \Delta x) \sin(k\pi x_j) - 2 \sin(k\pi x_j) \} \\ &= \underbrace{\frac{2}{\Delta x^2} \{ 1 - \cos(k\pi \Delta x) \}}_{\hat{\lambda}^k} \underbrace{\sin(k\pi x_j)}_{\hat{u}_j^k} . \end{aligned}$$

$$A \underline{\hat{u}}^k = \hat{\lambda}^k \underline{\hat{u}}^k$$

The Eigenvalue Problem

Error Analysis

...Analytical Solution: $\hat{u}^k, \hat{\lambda}^k$...



The Eigenvalue Problem

Error Analysis

Conclusions...

Low modes

For fixed k , $\Delta x \rightarrow 0$:

$$\begin{aligned}\hat{\lambda}^k &= \frac{2}{\Delta x^2} \{1 - \cos(k\pi \Delta x)\} \\ &= \frac{2}{\Delta x^2} \left\{1 - \left(1 - \frac{1}{2}k^2\pi^2\Delta x^2 + \mathcal{O}(\Delta x^4)\right)\right\} \\ &= k^2\pi^2 + \mathcal{O}(\Delta x^2)\end{aligned}$$

second-order convergence, $\hat{\lambda}^k \rightarrow \lambda^k$.

The Eigenvalue Problem

Error Analysis

...Conclusions...

High modes:

For $k = n$,

$$\Delta x = \frac{1}{n+1}$$

$$\begin{aligned}\hat{\lambda}^n &= \frac{2}{\Delta x^2} \left\{ 1 - \cos\left(\frac{n\pi}{n+1}\right) \right\} \\ &= 4(n+1)^2 \quad \text{as } \Delta x \rightarrow 0 \\ &\neq n^2\pi^2 = \lambda^n.\end{aligned}$$

High modes ($k \approx n$) **are not** accurate.

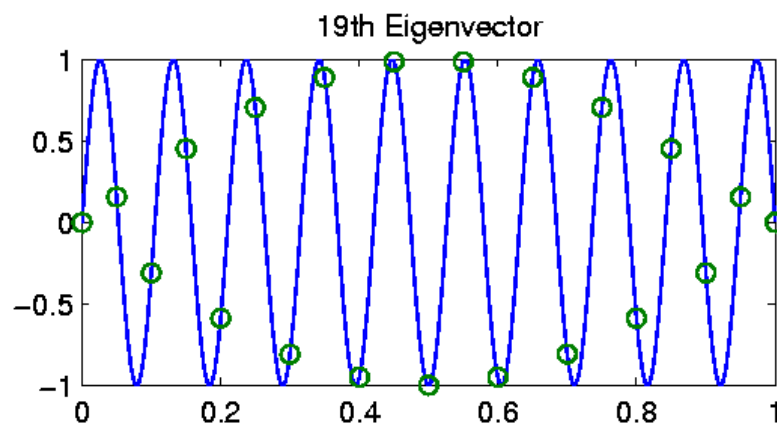
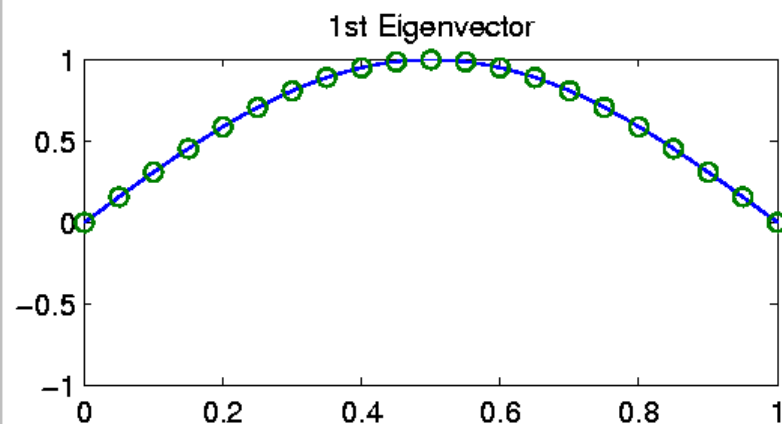
The Eigenvalue Problem

Error Analysis

...Conclusions...

Low modes vs. high modes

Example : $n = 19$, $\Delta x = 1/20$



The Eigenvalue Problem

Error Analysis

...Conclusions...

Low modes vs. high modes

$$k \ll n$$

$$k \approx n$$

N19

$\hat{u}^k$ resolved

$\hat{\lambda}^k$ accurate

$$\hat{\lambda}^k - \lambda^k \sim \mathcal{O}(\Delta x^2)$$

$\hat{u}^k$ not resolved

$\hat{\lambda}^k$ not accurate

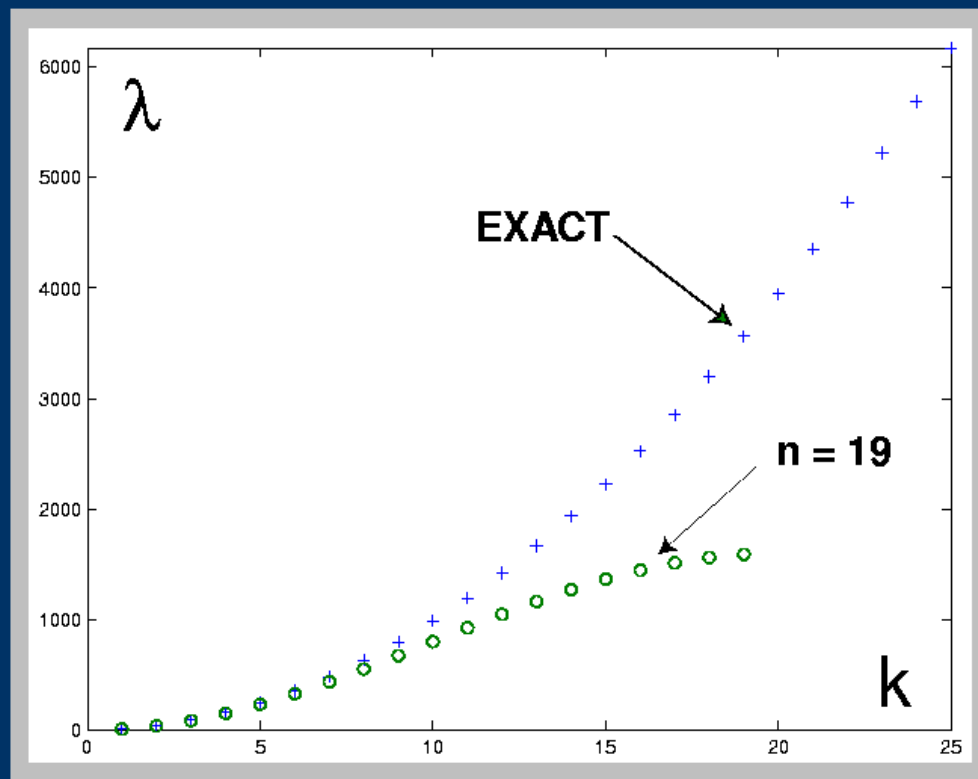
$$\hat{\lambda}^k - \lambda^k \text{ is } \mathcal{O}(1)$$

BUT: as $\Delta x \rightarrow 0$, $n \rightarrow \infty$, so any fixed mode k converges.

The Eigenvalue Problem

Error Analysis

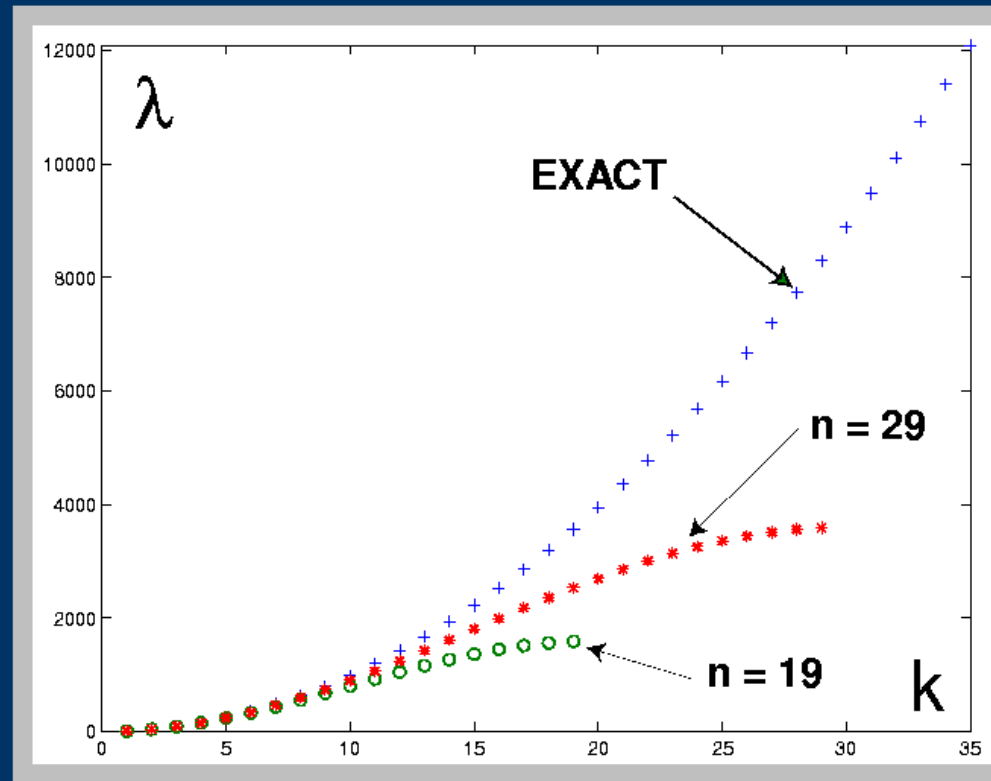
...Conclusions...



The Eigenvalue Problem

Error Analysis

...Conclusions



The Eigenvalue Problem

Condition Number of A

For a SPD matrix M , the condition number κ_M is given by

$$\kappa_M = \frac{\text{maximum eigenvalue of } M}{\text{minimum eigenvalue of } M}.$$

Thus, for our A matrix,

$$\kappa_A \rightarrow \frac{4n^2}{\pi^2} \text{ as } \Delta x \rightarrow 0$$

grows (in $\mathbb{R}^1$) as number of grid points squared. **N20**

Importance: understanding solution procedures.

Eigenvalue Problem in 2D

Statement

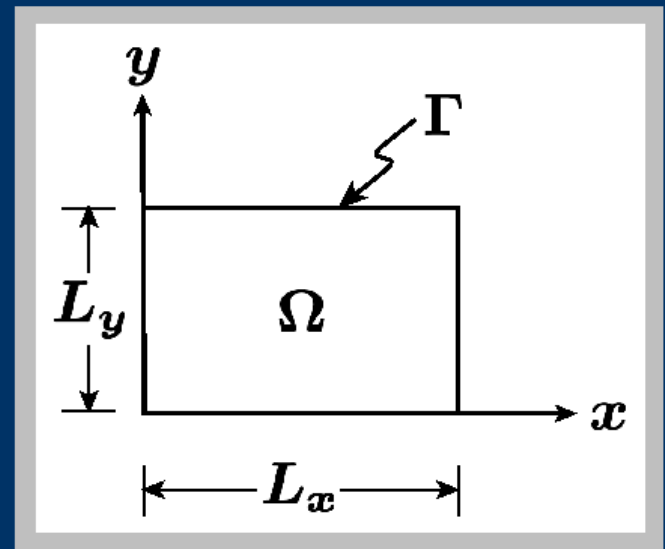
$$-\nabla^2 u = \lambda u \quad \text{in } \Omega$$

$$u = 0 \quad \text{on } \Gamma$$

Assume (for simplicity)

$$L_x = L_y = 1$$

Solutions $(u(x, y), \lambda)$



Eigenvalue Problem in 2D

Eigenvalues

$$u^{k,l}(x, y) = \sin(k\pi x) \sin(l\pi y)$$

$$-\nabla^2 u^{k,l} = (k^2\pi^2 + l^2\pi^2) u^{k,l}$$

Eigenvectors

$$\lambda^{k,l} = k^2\pi^2 + l^2\pi^2, \quad k, l = 1, \dots$$

Eigenvalue Problem in 2D

Discrete Problem

Eigenvectors

$$A\hat{u} = \hat{\lambda}\hat{u} \Rightarrow (\underline{u}^{k,l}, \hat{\lambda}^{k,l})$$

$$u_{i,j}^{k,l} = \sin(k\pi x_i) \sin(l\pi y_j)$$

$$= \sin(k\pi i\Delta x) \sin(l\pi j\Delta y)$$

$$= \sin\left(\frac{k\pi i}{n+1}\right) \sin\left(\frac{l\pi j}{m+1}\right)$$

$$k, i = 1, \dots, n \quad l, j = 1, \dots, m$$

Eigenvalue Problem in 2D

Discrete Problem

Eigenvalues

$$\hat{\lambda}^{k,l} = \frac{2}{\Delta x} \{1 - \cos(k\pi \Delta x)\} + \frac{2}{\Delta y} \{1 - \cos(l\pi \Delta y)\}$$

Low Modes

$\Delta x, \Delta y \rightarrow 0$ (k, l fixed)

$$\begin{aligned}\hat{\lambda}^{k,l} &= k^2\pi^2 + l^2\pi^2 \\ &+ \mathcal{O}(\Delta x^2, \Delta y^2)\end{aligned}$$

High Modes

$k \approx n, l \approx m$

$$\begin{aligned}\hat{\lambda}^{k,l} &= 4(n+1)^2 + 4(m+1)^2 \\ &\text{as } \Delta x, \Delta y \rightarrow 0\end{aligned}$$

Eigenvalue
Problem in 2D

$$\kappa_A \rightarrow \frac{4n^2 + 4m^2}{2\pi^2} \quad \text{as } \Delta x, \Delta y \rightarrow 0$$

If $m \approx n$

$$\kappa_A \rightarrow \frac{4n^2}{\pi^2}$$

grows (in $\mathbb{R}^2$) as number of grid points.
(better than in 1D, relatively speaking !!)